EE3621: Digital Signal Processing | III B.Tech EEE

Faculty Reference Document — Textbook Replacement

PREFACE FOR FACULTY

The Discrete Fourier Transform (DFT) is the cornerstone of practical digital signal processing. Unlike the DTFT (which produces a continuous spectrum) or the Z-Transform, the DFT operates on a finite sequence of N numbers and yields N discrete frequency coefficients, making it directly implementable in digital hardware and computers.

Pedagogical Strategy:

1. Derive the DFT by uniformly sampling the continuous DTFT spectrum $X(e^{j\omega})$ at N points $\omega_k = \frac{2\pi k}{N}$.
2. Introduce the twiddle factor notation $W_N = e^{-j2\pi/N}$ and prove its periodicity, symmetry, and orthogonality properties.
3. Formulate the DFT as a linear matrix transformation $\mathbf{X} = \mathbf{W}_N\mathbf{x}$.
4. Demonstrate the unitary matrix properties $\mathbf{W}_N^{-1} = \frac{1}{N}\mathbf{W}_N^*$.
5. Connect time-domain discrete resolution ($\Delta t = T_s$) to frequency-domain bin resolution ($\Delta f = F_s/N$).

1. LEARNING OBJECTIVES

By the end of this lecture, students will be able to:

1. **Derive** the forward N-point DFT and Inverse DFT (IDFT) analysis/synthesis equations.
2. **Construct** the $N \times N$ DFT twiddle factor matrix $\mathbf{W}_N$ for $N = 4$ and $N = 8$.
3. **Compute** the DFT of basic sequences analytically and via matrix-vector multiplication.
4. **Determine** frequency resolution Δf and bin frequencies $f_k = k\frac{F_s}{N}$.
5. **Relate** the DFT to the continuous DTFT and Discrete Fourier Series (DFS).

2. MATHEMATICAL FOUNDATIONS

2.1 The DFT Definition

For a finite-duration sequence $x[n]$ of length N ($0 \le n \le N-1$):

- **Forward N-point DFT:**

$$X[k] = \sum_{n=0}^{N-1} x[n]W_N^{nk}, \quad k = 0, 1, \dots, N-1$$

- **Inverse N-point IDFT:**

$$x[n] = \frac{1}{N}\sum_{k=0}^{N-1} X[k]W_N^{-nk}, \quad n = 0, 1, \dots, N-1$$

  Where the **Twiddle Factor** is defined as:

$$W_N = e^{-j\frac{2\pi}{N}} = \cos\left(\frac{2\pi}{N}\right) - j\sin\left(\frac{2\pi}{N}\right)$$

2.2 Fundamental Properties of the Twiddle Factor W_N

1. **Periodicity in n and k:**

$$W_N^{k+N} = W_N^k, \quad W_N^{nk+N} = W_N^{nk}$$

2. **Symmetry Property (Half-Period Inversion):**

$$W_N^{k+N/2} = -W_N^k \quad (\text{for even } N)$$

3. **Phase Reduction:**

$$W_N^{nk} = W_N^{(nk) \bmod N}$$

4. **Orthogonality of Basis Functions:**

$$\sum_{n=0}^{N-1} W_N^{n(k-m)} = \{N, \quad k = m \pmod N)\ 0, \quad k \neq m \pmod N)$$

2.3 Matrix Formulation of the DFT

In vector-matrix form:

$$\mathbf{X} = \mathbf{W}_N\mathbf{x}$$

$$\begin{bmatrix} X[0] & X[1] & X[2] & \vdots & X[N-1] \end{bmatrix} = \begin{bmatrix} W_N^0 & W_N^0 & W_N^0 & \dots & W_N^0 & W_N^0 & W_N^1 & W_N^2 & \dots & W_N^{N-1} & W_N^0 & W_N^2 & W_N^4 & \dots & W_N^{2(N-1)} & \vdots & \vdots & \vdots & \ddots & \vdots & W_N^0 \end{bmatrix}$$

The Inverse DFT is:

$$\mathbf{x} = \mathbf{W}_N^{-1}\mathbf{X} = \frac{1}{N}\mathbf{W}_N^*\mathbf{X}$$

Where $\mathbf{W}_N^*$ is the complex conjugate of the twiddle matrix.

For $N = 4$ ($W_4 = e^{-j2\pi/4} = e^{-j\pi/2} = -j$):

$$\mathbf{W}_4 = [1 \quad 1 \quad 1 \quad 1 \, 1 \quad -j \quad -1 \quad j \, 1 \quad -1 \quad 1 \quad -1 \, 1 \quad j \quad -1 \quad -j]$$

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3. WORKED NUMERICAL EXAMPLES

Example 7.1: 4-Point DFT Matrix Computation

Problem: Compute the 4-point DFT of $x[n] = \underset{\uparrow}{1}, 2, 0, 1$.

Solution: Using $\mathbf{X} = \mathbf{W}_4\mathbf{x}$:

$$[X[0] \; X[1] \; X[2] \; X[3]] = [1 \quad 1 \quad 1 \quad 1 \, 1 \quad -j \quad -1 \quad j \, 1 \quad -1 \quad 1 \quad -1 \, 1 \quad j \quad -1 \quad -j] \, [1 \; 2 \; 0 \; 1]$$

- $X[0] = 1(1) + 1(2) + 1(0) + 1(1) = 4$
- $X[1] = 1(1) - j(2) - 1(0) + j(1) = 1 - 2j + j = 1 - j$
- $X[2] = 1(1) - 1(2) + 1(0) - 1(1) = 1 - 2 - 1 = -2$
- $X[3] = 1(1) + j(2) - 1(0) - j(1) = 1 + 2j - j = 1 + j$ Result:

$$X[k] = 4, ; 1 - j, ; -2, ; 1 + j$$

Notice Hermitian symmetry: $X[0] = 4 \in \mathbb{R}$, $X[2] = -2 \in \mathbb{R}$, and $X[3] = X[1]^* = 1 + j$.

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4. UNIVERSITY EXAMINATION QUESTIONS & MARKING RUBRIC

Question 1 (15 Marks)

(a) State and prove the twiddle factor properties: Periodicity, Symmetry, and Orthogonality. *(6 Marks)* **(b)** Given an 8-point sequence $x[n] = \delta[n] + 2\delta[n-2] + \delta[n-4]$.

1. Compute its 8-point DFT $X[k]$ analytically. *(5 Marks)*
2. If sampling rate $F_s = 8000$ Hz, find the frequency in Hz corresponding to bin $k = 3$. *(2 Marks)*
3. Verify Parseval's relation for this sequence. *(2 Marks)*

Model Answer & Step-by-Step Marking Rubric:

- **Part (a):**
  - Periodicity proof: $W_N^{k+N} = e^{-j\frac{2\pi(k+N)}{N}} = e^{-j\frac{2\pi k}{N}}e^{-j2\pi} = W_N^k \cdot 1 = W_N^k$ *(2 Marks)*
  - Symmetry proof: $W_N^{k+N/2} = e^{-j\frac{2\pi(k+N/2)}{N}} = W_N^k e^{-j\pi} = -W_N^k$ *(2 Marks)*
  - Orthogonality proof: Geometric series $\sum_{n=0}^{N-1}[W_N^{k-m}]^n = \frac{1-W_N^{N(k-m)}}{1-W_N^{k-m}} = \frac{1-1}{1-W_N^{k-m}} = 0$ for $k \neq m$ *(2 Marks)*
- **Part (b.1):**
  - $X[k] = \sum_{n=0}^{7} x[n]W_8^{nk} = 1 \cdot W_8^0 + 2W_8^{2k} + 1 \cdot W_8^{4k}$
  - Since $W_8^2 = W_4 = e^{-j\pi/2} = -j$ and $W_8^4 = W_2 = e^{-j\pi} = -1$:

    $$X[k] = 1 + 2(-j)^k + (-1)^k$$

    - $X[0] = 1 + 2(1) + 1 = 4$
    - $X[1] = 1 + 2(-j) - 1 = -2j$
    - $X[2] = 1 + 2(-1) + 1 = 0$
    - $X[3] = 1 + 2(j) - 1 = 2j$
    - $X[4] = 1 + 2(1) + 1 = 4$
    - $X[5] = 1 + 2(-j) - 1 = -2j$
    - $X[6] = 1 + 2(-1) + 1 = 0$
    - $X[7] = 1 + 2(j) - 1 = 2j$

    $$X[k] = 4, -2j, 0, 2j, 4, -2j, 0, 2j$$

    (5 Marks)
- **Part (b.2):**
  - Bin frequency: $f_k = k\frac{F_s}{N} = 3 \times \frac{8000}{8} = 3000$ Hz. *(2 Marks)*
- **Part (b.3):**
  - Time energy: $\sum_{n=0}^{7} |x[n]|^2 = 1^2 + 2^2 + 1^2 = 1 + 4 + 1 = 6$.

- Frequency energy: $\frac{1}{8}\sum_{k=0}^{7}|X[k]|^2 = \frac{1}{8}[4^2+2^2+0+2^2+4^2+2^2+0+2^2] = \frac{1}{8}[16+4+4+16+4+4] = \frac{48}{8} = 6$.
- Parseval's holds exactly $(6 = 6)$. *(2 Marks)*

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5. PYTHON VERIFICATION SCRIPT

```
import numpy as np

x = np.array([1, 0, 2, 0, 1, 0, 0, 0])
X = np.fft.fft(x)
print("Computed 8-pt DFT:", np.round(X, 4))
```